

T-REX: A Dual-Mode DiskSat Constellation

Dispersed VLBI vs. Combined Single-Dish Sensitivity

T-REX Team

The T-REX architecture supports two complementary observing modes enabled by the same 18 hexagonal DiskSats. In *dispersed mode*, the DiskSats separate to fill a wide-field constellation, each operating as an independent 86 GHz station in a space-VLBI array. This maximizes angular resolution, since baseline lengths scale with DiskSat separation rather than aperture size, but each element has a small collecting area and thus a high system equivalent flux density (SEFD), limiting sensitivity to bright, compact sources such as Sgr A* and M87*. In *combined mode*, the DiskSats dock and interlock into a single monolithic mirror, pooling their collecting area into one large aperture. This sacrifices the long baselines (and thus angular resolution) in exchange for a dramatic gain in point-source sensitivity, useful for flux calibration, weak or newly triggered transient targets, and blind time-domain surveys where sub-microarcsecond resolution is not required.

Each hexagonal segment has circumradius $R = 0.715$ m, giving a per-segment area $A_1 = \frac{3\sqrt{3}}{2}R^2 \approx 1.33$ m²; the 18-segment combined aperture totals $A_{18} \approx 23.9$ m². Using the radiometer relation

$$\text{SEFD} = \frac{2k_B T_{\text{sys}}}{\eta A},$$

with $T_{\text{sys}} = 150$ K and aperture efficiency $\eta = 0.65$ (typical of an 86 GHz receiver chain), a single DiskSat has $\text{SEFD}_1 \approx 4.8 \times 10^5$ Jy, while the fully combined mirror achieves $\text{SEFD}_{18} \approx 2.7 \times 10^4$ Jy — an improvement of exactly a factor of 18, since $\text{SEFD} \propto 1/A$ and $A_{18} = 18 A_1$. Equivalently, combined mode reaches the same signal-to-noise ratio as a single DiskSat in 1/18 of the integration time, making the two modes directly tunable against the resolution–sensitivity trade central to T-REX’s science case.